



Ministry of Education
SINGAPORE



Scoring & Strategies

How scoring works

You team are given scores based on the following. Let's have a look:

Meeting Scenario Objectives:

Some scenarios will give you points meeting their requirements. For instance, in the scenario, "Burger Jam", 50 points are awarded for each burger that leaves the stomach.

Saliva Application:

Points are awarded according to the how much time is left on the **sequence countdown timer**, and how much saliva is applied.

To earn most points, **apply the saliva completely, and enter the sequence correctly in the shortest possible time**. Entering the sequence correctly will help restore some time to the countdown, while entering the sequence wrongly will shorten the timer. Do not cheat by leaving the hotseat to reset the countdown timer. Scoring for each food is based only on the first time you operate the hotseat for that food.

Leaving the hotseat in the middle of a sequence will give you a lower score

If a food containing carbohydrates is chosen, points are awarded for starting carbohydrate digestion. 1 point is awarded for applying saliva on an exposed carbohydrate (shown in red circle with number in it).

Mastication(biting and chewing):

You will have points awarded or docked based on the hardness level remaining at the point that you swallow the food. Normally, you should be able to get full points for this. The question is whether you can do so with a minimum amount of bites and quick timing.

hardness left	0 – 9	10 – 19	20 – 29	30 - 39	40 – 49	50 -59	60 – 69	70 -79	80 – 89	90 – 99	100
score	5	4	3	2	1	0	-1	-2	-3	-4	-5

If timing is an important consideration for the scenario, consider applying saliva and chewing at the same time. The effect of saliva on chewing strength is as such:

Saliva at 50% or less – Chewing strength halved

Saliva greater than 50% and less than 100% - Saliva percentage X Chewing strength

Saliva at 100% - Full Chewing Strength.

Try aiming for the molars for maximum effect. Also try alternating between the left and right molars if you need to get a quick timing.

Oesophagus, peristalsis points:

5 points are awarded every time perfect peristalsis happens, regardless of how many boli there are in the passage. In addition, the higher the speed of the bolus, the faster you recover after each round of peristalsis.

Hydrochloric Acid pH Management Incentive:

Points are awarded for successful release of Hcl acid using the following scoring system:

Time taken(sec)	1 to 7	8 to 15	16 to 25	> 25
score	8	4	1	0

Stomach, peristalsis and pepsin points:

Points at this control is based on the outcome displayed across the three slots:

15 points perfect churn	All three slots at centre row are identical symbols.
10 points good churn	Two slots at centre row are identical symbols, one slot shows the symbol off centre.
5 points average churn	Two slots at centre row are identical symbols, one slot doesn't show the symbol at all.
5 points average churn	Symbol at centre row of first slot is found in other two slots, but both symbols are off centre.
0 points bad churn	Symbol at centre row of first slot is not found in the other two slots at all.

Protein digestion by pepsin will not earn you any points but will make the task at turret 1 easier.

Accuracy allowance and Enzyme conservation scheme:

For successfully digesting a food molecule into a simpler form:
you are awarded 5 points.

For emulsifying fat with bile for easier digestion:
you are awarded 5 points.

As part of our enzyme conservation scheme, you will be penalised 1 point for every enzyme missed or for matching the wrong enzyme to the food. Make every shot count.

Nutrient Absorption and Sorting:

Most of your score would come from this:

- +10 points for absorption of fully digested nutrient into **correct pipe**
- +1 point for absorption of fully digested nutrient into **wrong pipe**. Lower score due to additional sorting used to correct error.
- -5 points for waste of fully digested nutrient (passing to colon unabsorbed)
- -5 points for attempting to absorb fibre.
- -5 points for waste of undigested or partially undigested nutrient (passing to colon unabsorbed)
- -10 points for attempting to absorb undigested food, partially digested food (includes food wastage penalty)

Crisis costs and other penalties

Crisis between the Mouth & Oesophagus:

The GUT REPLACEMENT MODULE will shutdown when there are more than 2 boli before the oesophagus. 1 point will be deducted for every second until the queue problem is fixed.

Crisis between the Oesophagus and Stomach:

The GUT REPLACEMENT MODULE will shutdown when there are more than 2 boli before the stomach. 1 point will be deducted for every second until the queue problem is fixed.

Food wastage

Digestico strongly discourages the wasting of food. Food that is undigested or partially digested cannot be absorbed and will be considered as wasted food. To prevent waste, choose your meals carefully.

Hotseat penalty

A penalty of 5 points is imposed for each second when you exceed the allocated 120secs of usage. The point deduction stops when you get up from the hotseat.

NUTRITIONAL INFORMATION

Nutrients that automatons see are in the form of blocks. One block holds 3g of a nutrient.

1g of protein will yield	4 CALORIES(kcal) of energy
1g of carbohydrate will yield	4 CALORIES(kcal) of energy
1g of fat will yield	9 CALORIES (kcal) of energy.

The normal human require about 2000-2200 CALORIES(or kcal) of food intake a day . George's ideal intake is ~2100 CALORIES a day. His daily recommended nutrition in blocks is :
96 carbohydrate: 24 fat: 24 protein

Try to keep within reasonable limits of the recommended intake to keep George in good health.
ie not less than 1622 CALORIES DAILY & not more than 2359 CALORIES DAILY
Data of George's food diet is in the following pages for your reference.

Food Data

Baked Beans



18% 80% 2%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	1	3g
Carbohydrate	3	9g
Fat	0	0g
Fibre	1	3g
Water	11	33g
Total:	16	48g
Hardness Value: 450		

A good source of dietary fiber, for healthy bowel function.

Curry Chicken



69% 1% 30%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	17	51g
Carbohydrate	0	0g
Fat	4	12g
Fibre	0	0g
Water	43	129g
Total:	64	192g
Hardness Value: 950		

Chicken seasoned with thick curry gravy. It makes a tasty meal.

Mixed Vegetables Dish



12% 86% 2%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	2	6g
Carbohydrate	8	24g
Fat	0	0g
Fibre	3	9g
Water	43	129g
Total:	51	153g
Hardness Value: 900		

For the health-conscious, this dish provides the necessary nutrition your body needs.

Steamed Snapper Fish



88% 0% 12%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	17	51g
Carbohydrate	0	0g
Fat	1	3g
Fibre	0	0g
Water	46	138g
Total:	64	192g
Hardness Value: 600		

An excellent source of omega 3 fatty acids. It improves the health of our brain and eyes.

Fruity Muesli Bar



11% 76% 13%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	2	6g
Carbohydrate	12	36g
Fat	1	3g
Fibre	1	3g
Water	0	0g
Total:	16	48g
Hardness Value: 850		

Yummy and nutritious light snack made up of whole grains and various nuts.

Tropical Jawbreaker



0% 100% 0%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	0	0g
Carbohydrate	8	24g
Fat	0	0g
Fibre	0	0g
Water	0	0g
Total:	8	24g
Hardness Value: 1000		

Delightful sweets to satisfy your sweet tooth craving. Keeps your mouth busy all day.

Fried Egg



35% 3% 62%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	2	6g
Carbohydrate	0	0g
Fat	2	6g
Fibre	0	0g
Water	12	36g
Total:	16	48g
Hardness Value: 400		

Provide nutrients that help regulate cholesterol level and develop muscles.

Chocolate Bar



4% 40% 56%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	1	3g
Carbohydrate	9	27g
Fat	4	12g
Fibre	1	3g
Water	1	3g
Total:	16	48g
Hardness Value: 800		

Has antioxidants (flavonoids) which help protect against free radicals that cause cell and tissue damage.

Banana



4% 93% 3%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	0	0g
Carbohydrate	8	24g
Fat	0	0g
Fibre	1	3g
Water	23	69g
Total:	16	48g
Hardness Value: 400		

An excellent source of potassium and fibre which helps in our digestion.

Red Apple



0% 100% 0%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	0	0g
Carbohydrate	1	3g
Fat	0	0g
Fibre	1	3g
Water	6	18g
Total:	8	24g
Hardness Value: 800		

An excellent source of antioxidants and able to relieve constipation.

Cookie Chips



5% 65% 30%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	1	3g
Carbohydrate	5	15g
Fat	2	6g
Fibre	0	0g
Water	0	0g
Total:	8	24g
Hardness Value: 600		

Huge crunchy cookies that makes a worthy light-hearted snack.

French Fries



4% 48% 48%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	1	3g
Carbohydrate	12	36g
Fat	6	18g
Fibre	1	3g
Water	12	36g
Total:	32	96g
Hardness Value: 300		

Looking for a quick snack? This crispy golden fries is just what you need!

Bowl of Rice



7% 92% 7%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	4	12g
Carbohydrate	51	153g
Fat	0	0g
Fibre	1	3g
Water	8	24g
Total:	64	192g
Hardness Value: 600		

Our main staple of food. Has great source of complex carbohydrates, to give our body energy.

Dried Noodles



6% 45% 49%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	2	6g
Carbohydrate	18	54g
Fat	9	27g
Fibre	1	3g
Water	2	6g
Total:	32	96g
Hardness Value: 600		

It is filling and easy to digest. However they are low in other nutrients.

Strawberry Yogurt



16% 77% 7%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	1	3g
Carbohydrate	6	18g
Fat	0	0g
Fibre	0	0g
Water	25	75g
Total:	32	96g
Hardness Value: 100		

Makes a healthy and tasty dessert!

Cheese Burger



23% 42% 35%
Protein Carbo Fats

Nutrition Facts

	Blocks	Qty
Protein	8	24g
Carbohydrate	19	27g
Fat	8	24g
Fibre	1	3g
Water	28	84g
Total:	64	192g
Hardness Value: 1000		

Tasty but has a high amount of carbohydrate and calories enough to fill an empty stomach.